

Customer Churn Prediction – Model Evaluation Report

Executive Summary

This report presents the evaluation results of multiple machine learning models developed to predict customer churn. The primary business objective is to accurately identify customers at risk of churning, enabling proactive retention strategies. Given the imbalance in churn data, evaluation focused on Recall, F1-score, ROC-AUC, and Precision-Recall AUC rather than accuracy alone.

Models Evaluated

- Logistic Regression
- Random Forest Classifier
- XGBoost Classifier

Model Performance Summary (Test Set)

Model	ROC-AUC	PR-AUC	Recall (Churn)	F1-score (Churn)
Logistic Regression	0.829	0.567	0.72	0.59
Random Forest	0.843	0.620	0.71	0.61
XGBoost	0.832	0.623	0.65	0.59

Business Interpretation

- **Random Forest** demonstrated the best overall balance between identifying churners and limiting false positives. It achieved the highest ROC-AUC and F1-score, making it the most reliable model for operational deployment.
- **XGBoost** achieved the highest PR-AUC, indicating strong performance in ranking high-risk customers, but slightly lower recall makes it less effective at capturing all churners.
- **Logistic Regression** provided strong recall but generated a higher number of false positives, which may increase unnecessary retention costs.

Final Recommendation

The Random Forest model is recommended for deployment in the churn prediction system. It provides the best trade-off between churn detection and business efficiency. Further improvements can be achieved through probability threshold tuning to align model behavior with specific business retention strategies.

Telco Customer Churn Analysis Report

Summary and Key Findings:

Summary:

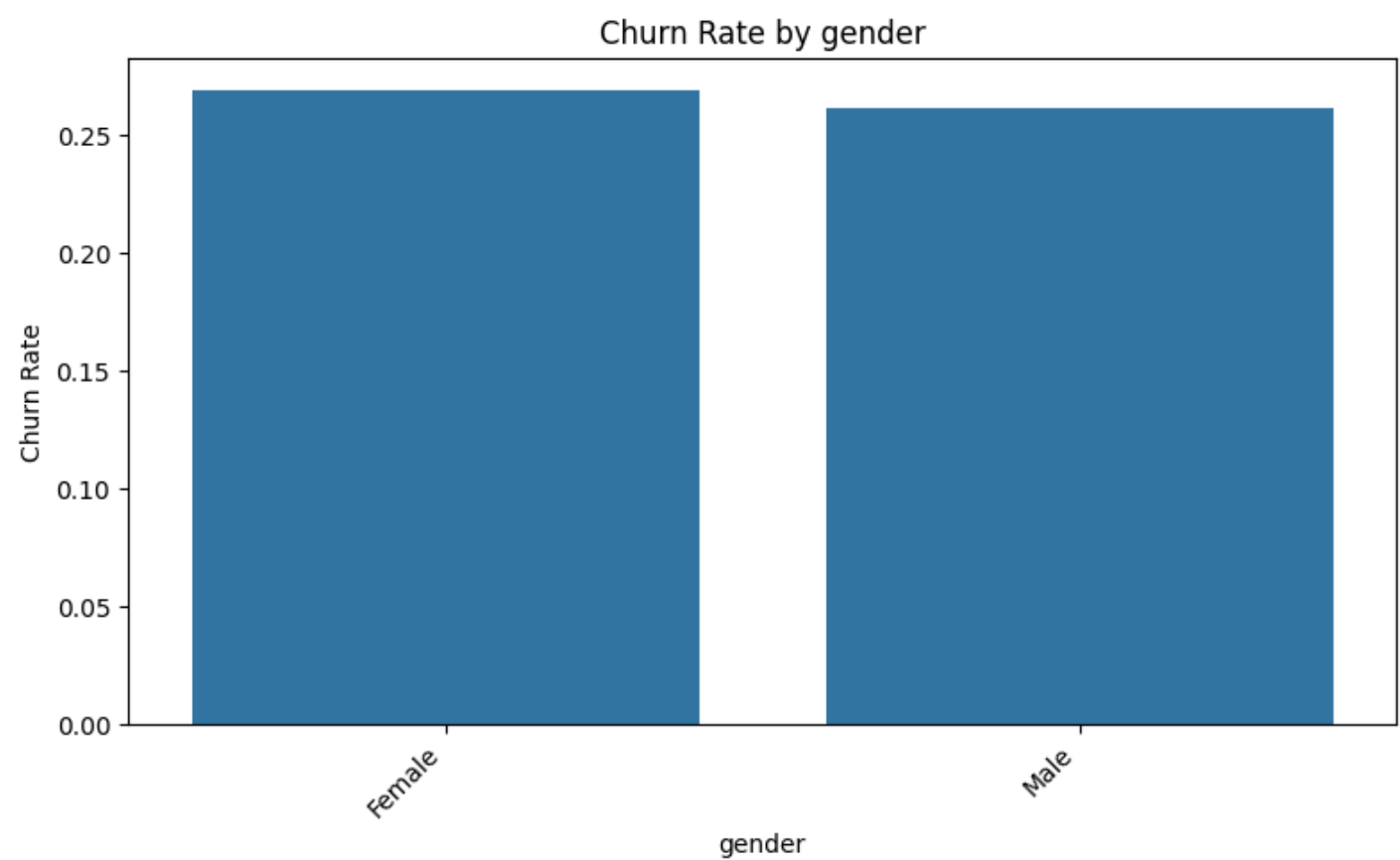
Data Analysis Key Findings

- * **Categorical Churn Rates:** Churn rates were calculated and visualized for 16 categorical features including `gender`, `SeniorCitizen`, `Partner`, `Dependents`, `PhoneService`, `MultipleLines`, `InternetService`, `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, `StreamingMovies`, `Contract`, `PaperlessBilling`, and `PaymentMethod`. The visualizations provide insights into how different categories within these features correlate with customer churn.
- * **Numerical Feature Distribution vs. Churn:** Violin plots were used to visualize the distribution of `tenure`, `MonthlyCharges`, and `TotalCharges` against customer churn, illustrating differences in these numerical features between churning and non-churning customers.
- * **XGBoost Feature Importance:** The XGBoost model identified 'Contract' as the most important feature with an importance score of approximately 0.41. The top 10 most influential features were extracted and visualized, providing a clear ranking of feature relevance according to the model.

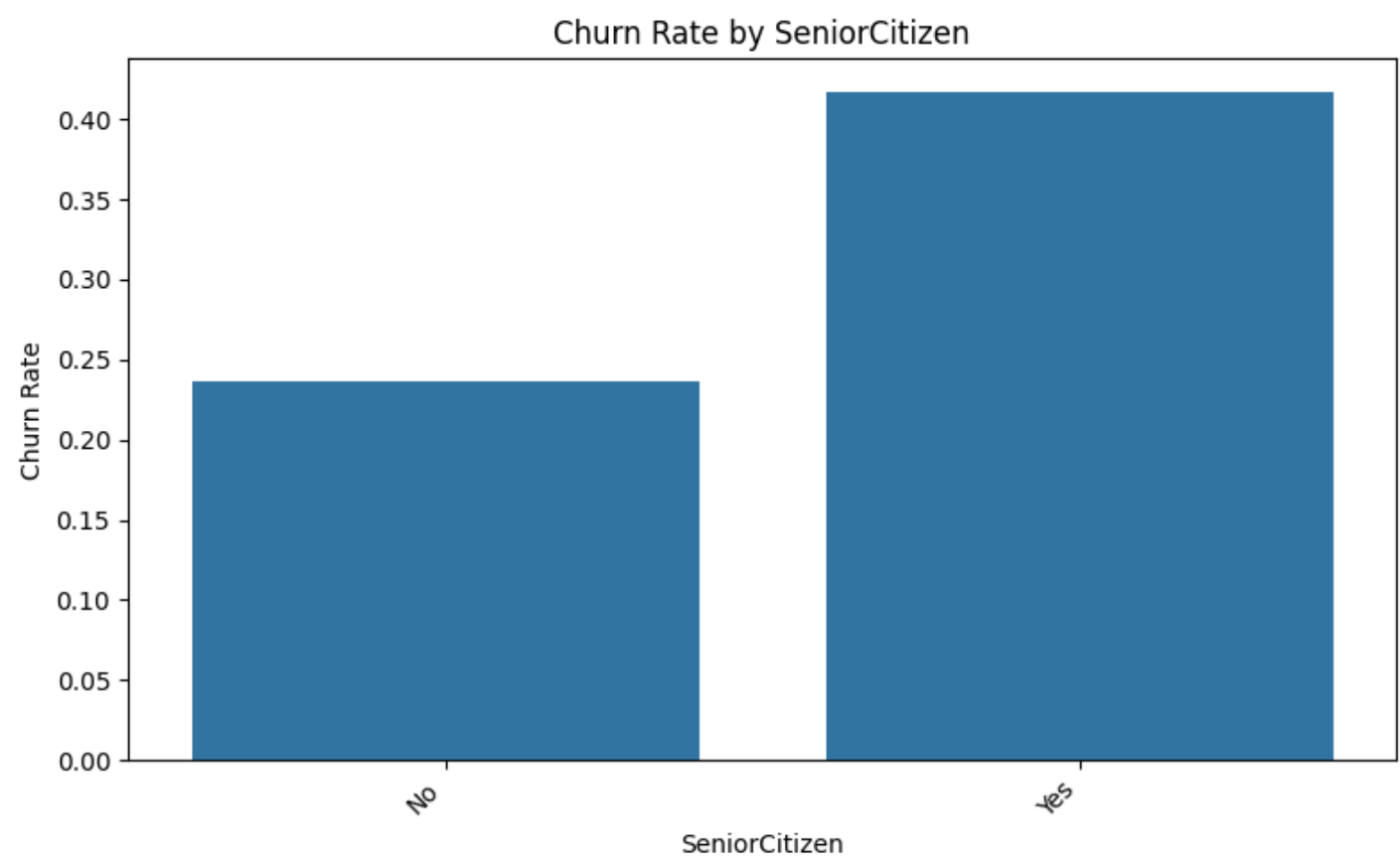
Insights or Next Steps

- * **Targeted Retention Strategies:** The high importance of 'Contract' and the churn rates observed for specific categories within other features (e.g., 'InternetService', 'PaymentMethod' if specific insights were evident from the plots) suggest that retention efforts could be targeted based on these attributes, such as offering incentives for longer-term contracts or improving services related to high-churn categories.
- * **Further Feature Exploration:** Given 'Contract's' significant importance, further analysis could delve deeper into the types of contracts (e.g., month-to-month vs. annual) and their specific impact on churn, potentially leading to actionable business strategies.

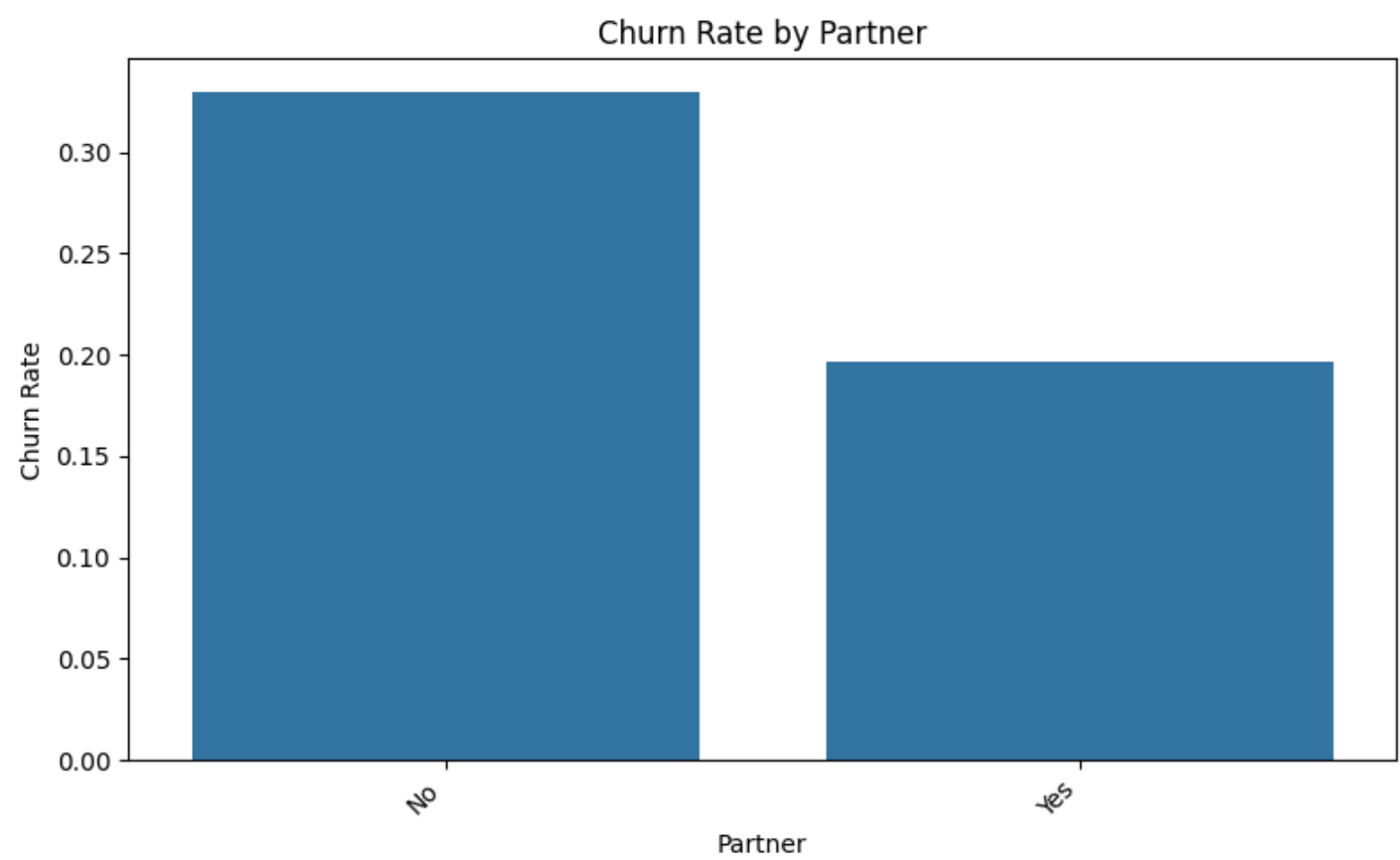
Churn Rate by gender



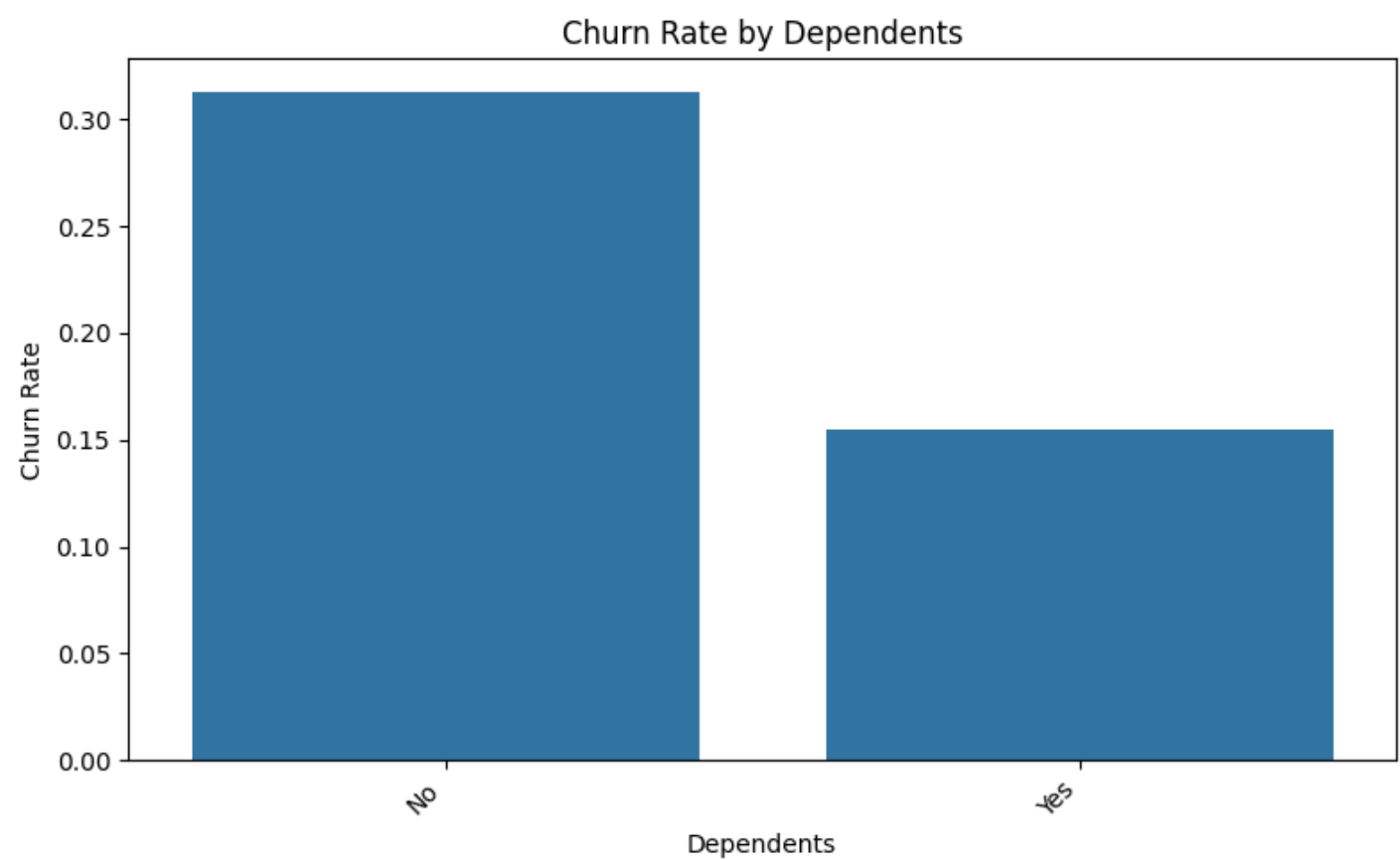
Churn Rate by SeniorCitizen



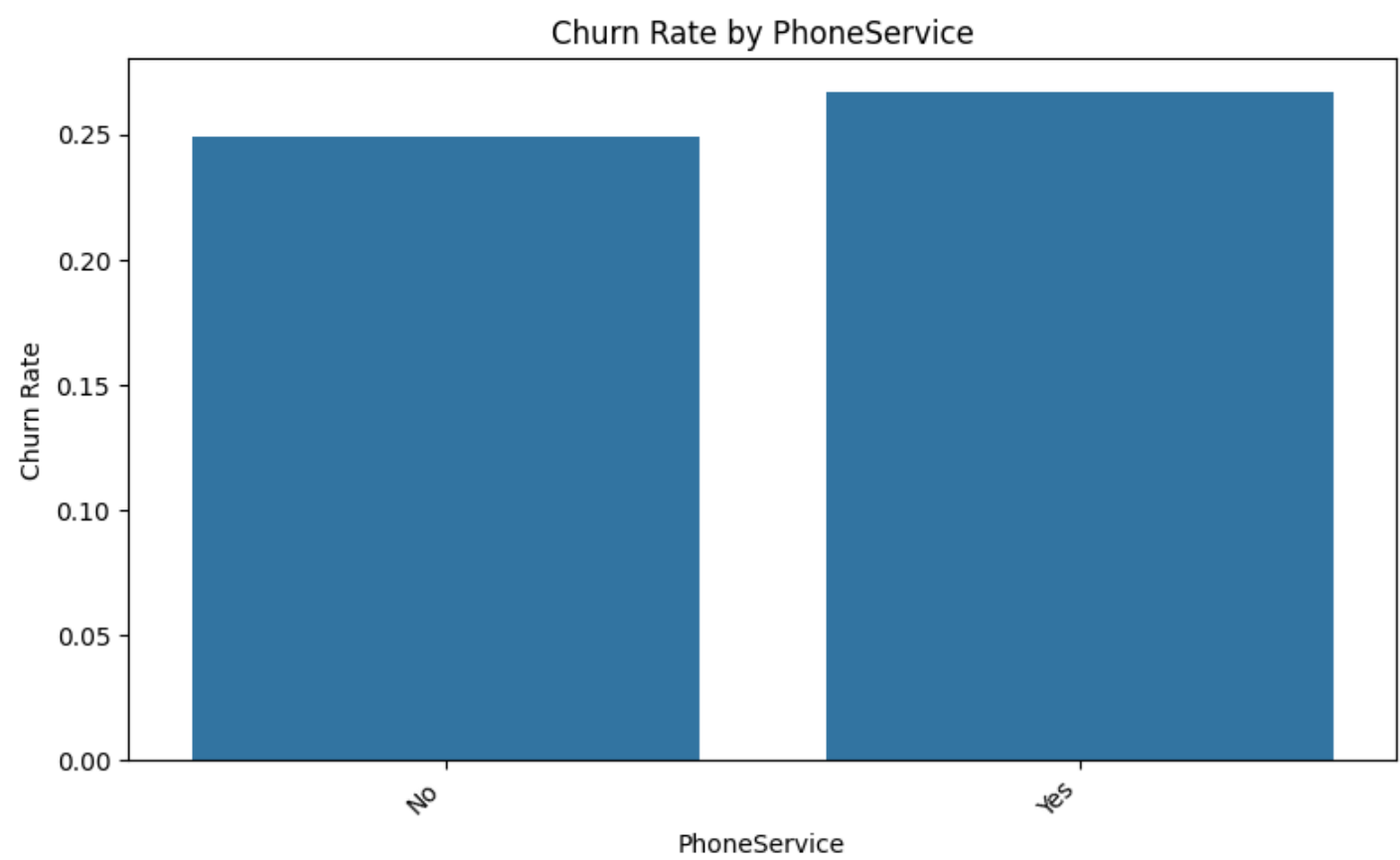
Churn Rate by Partner



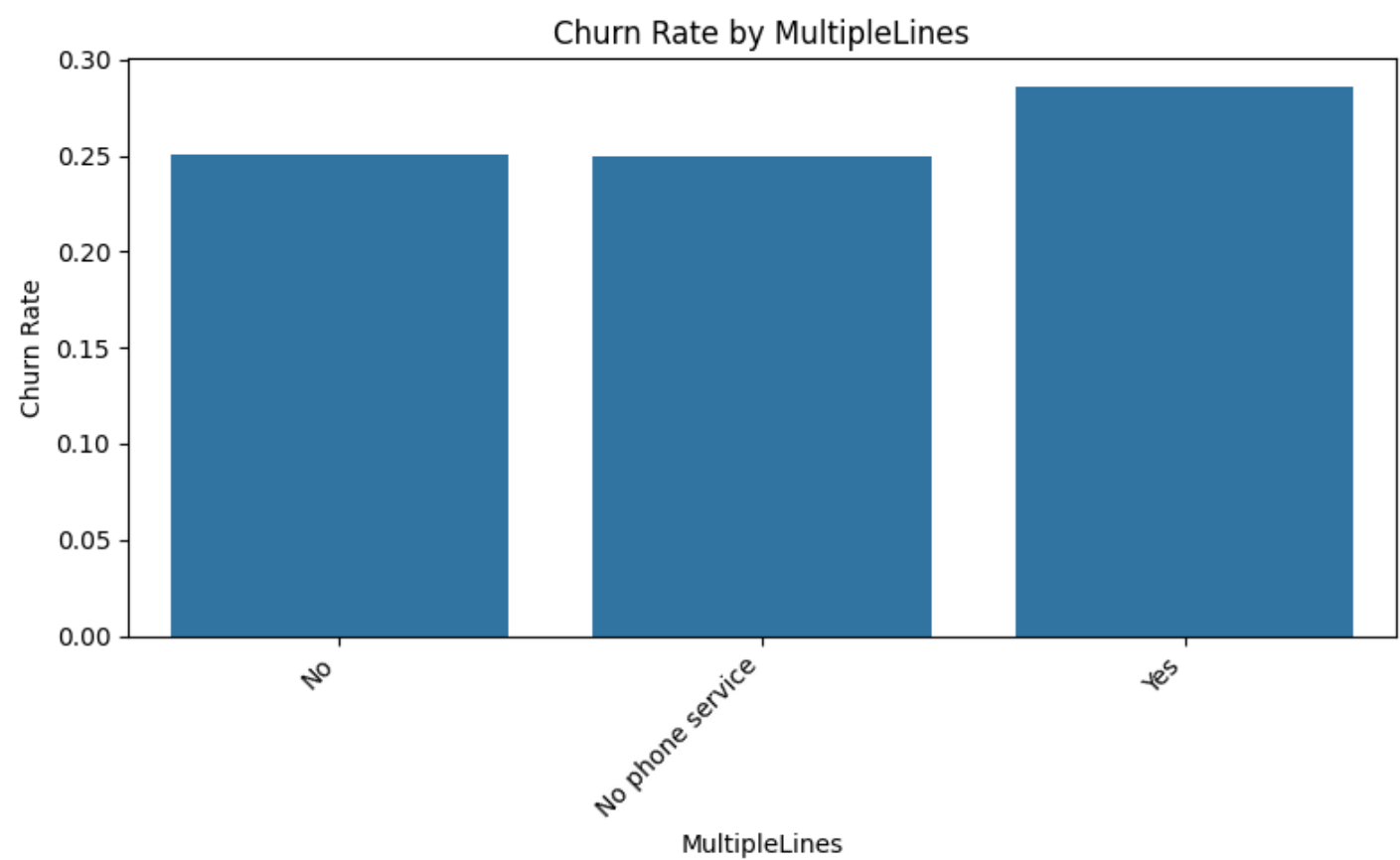
Churn Rate by Dependents



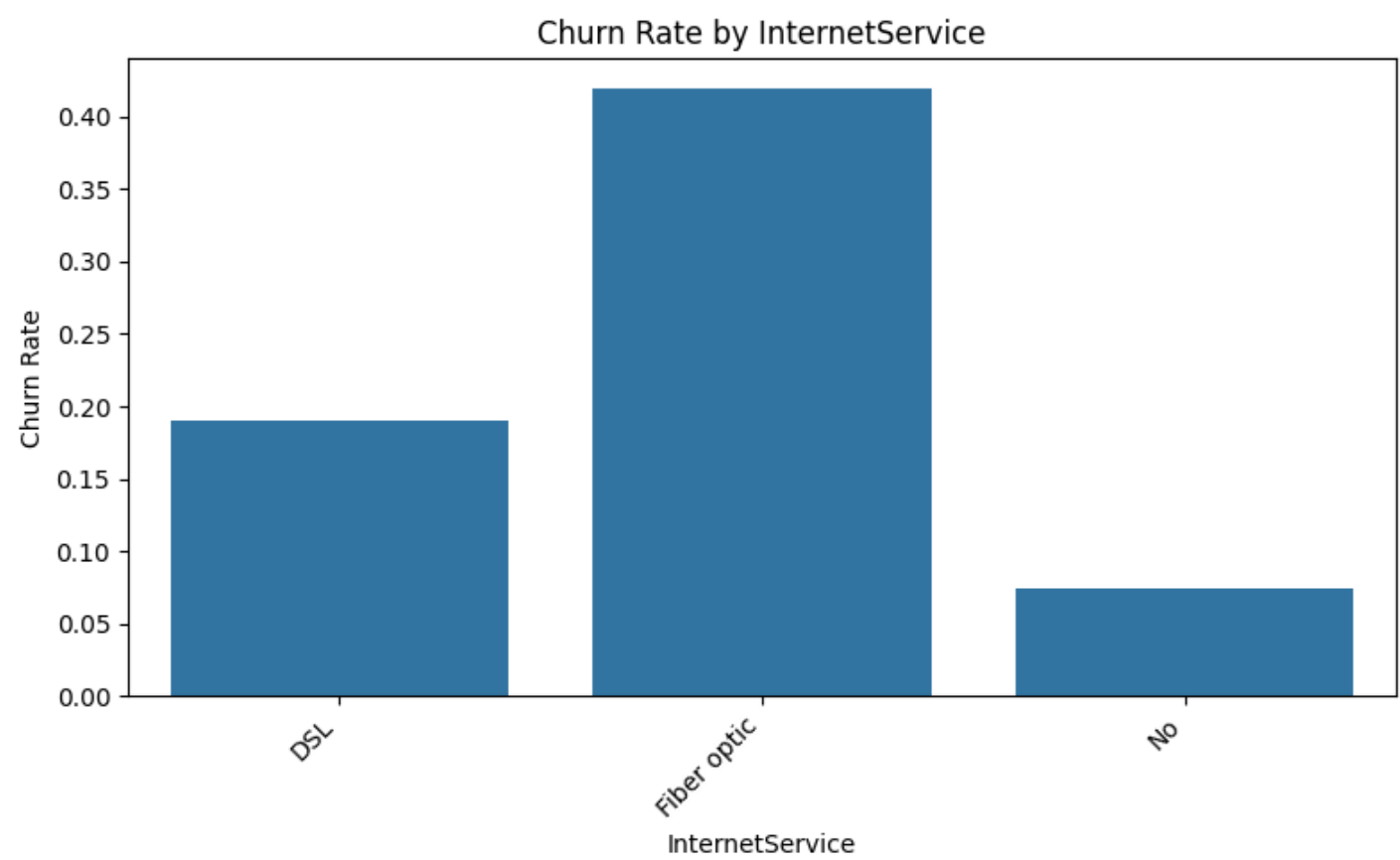
Churn Rate by PhoneService



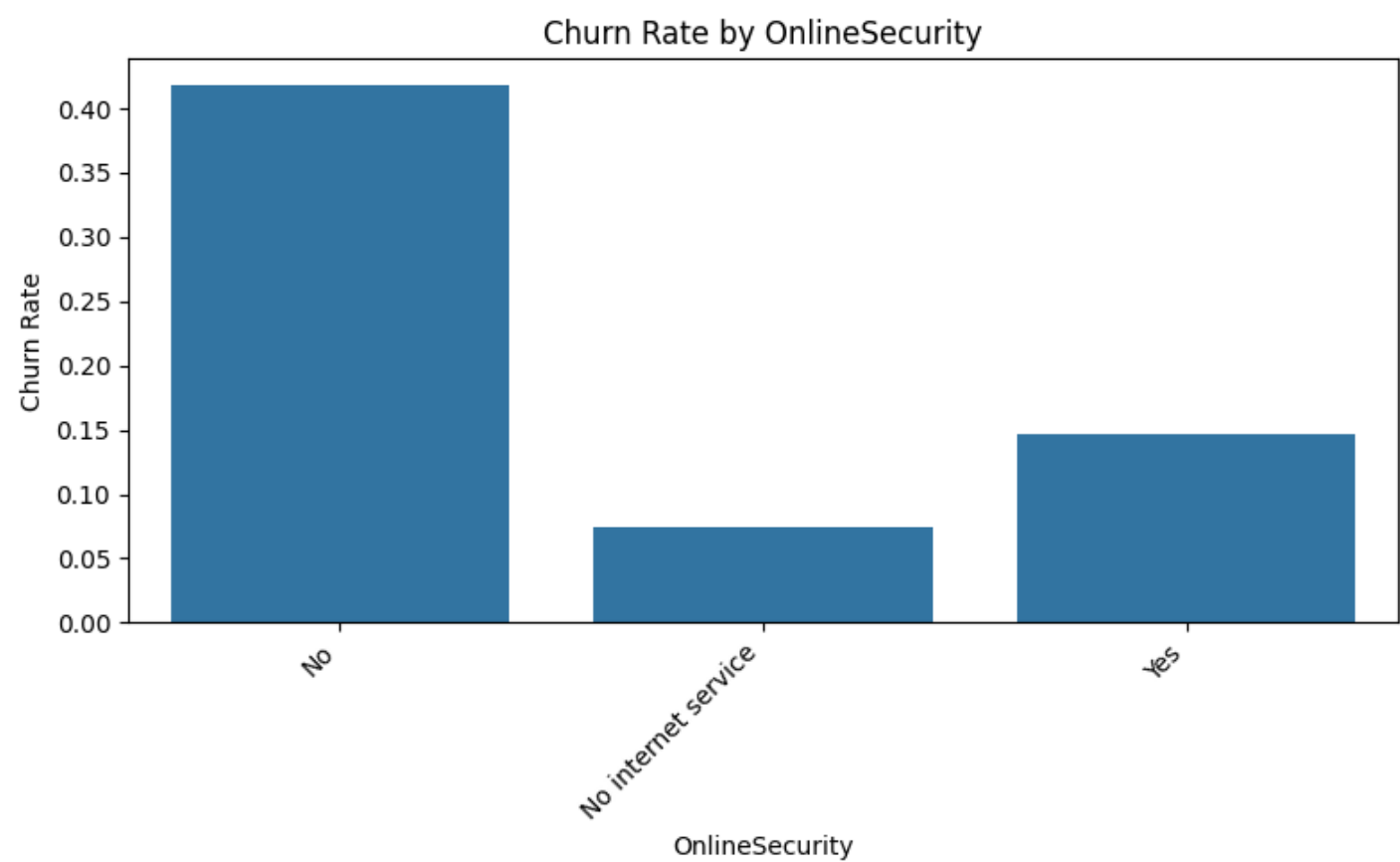
Churn Rate by MultipleLines



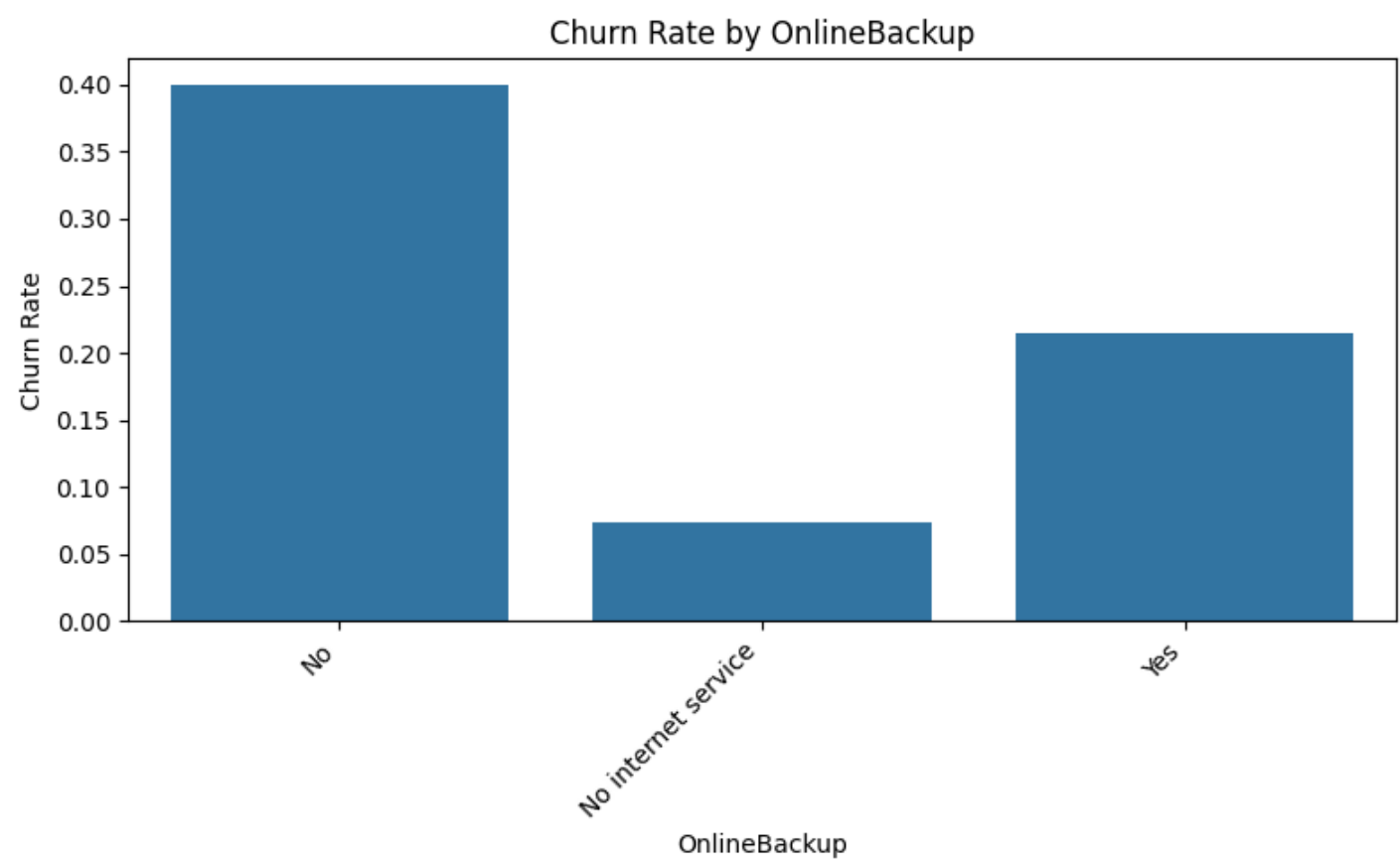
Churn Rate by InternetService



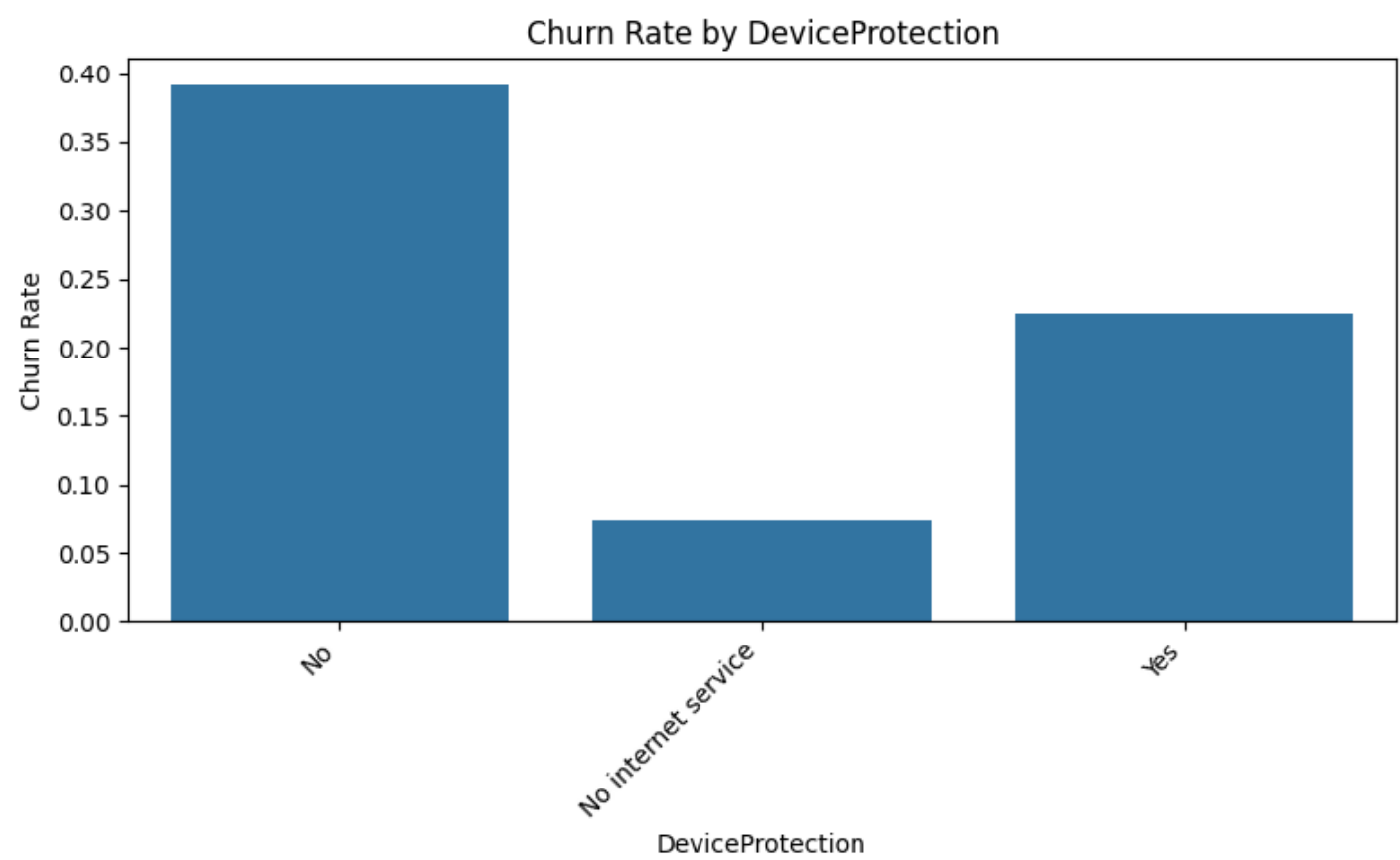
Churn Rate by OnlineSecurity



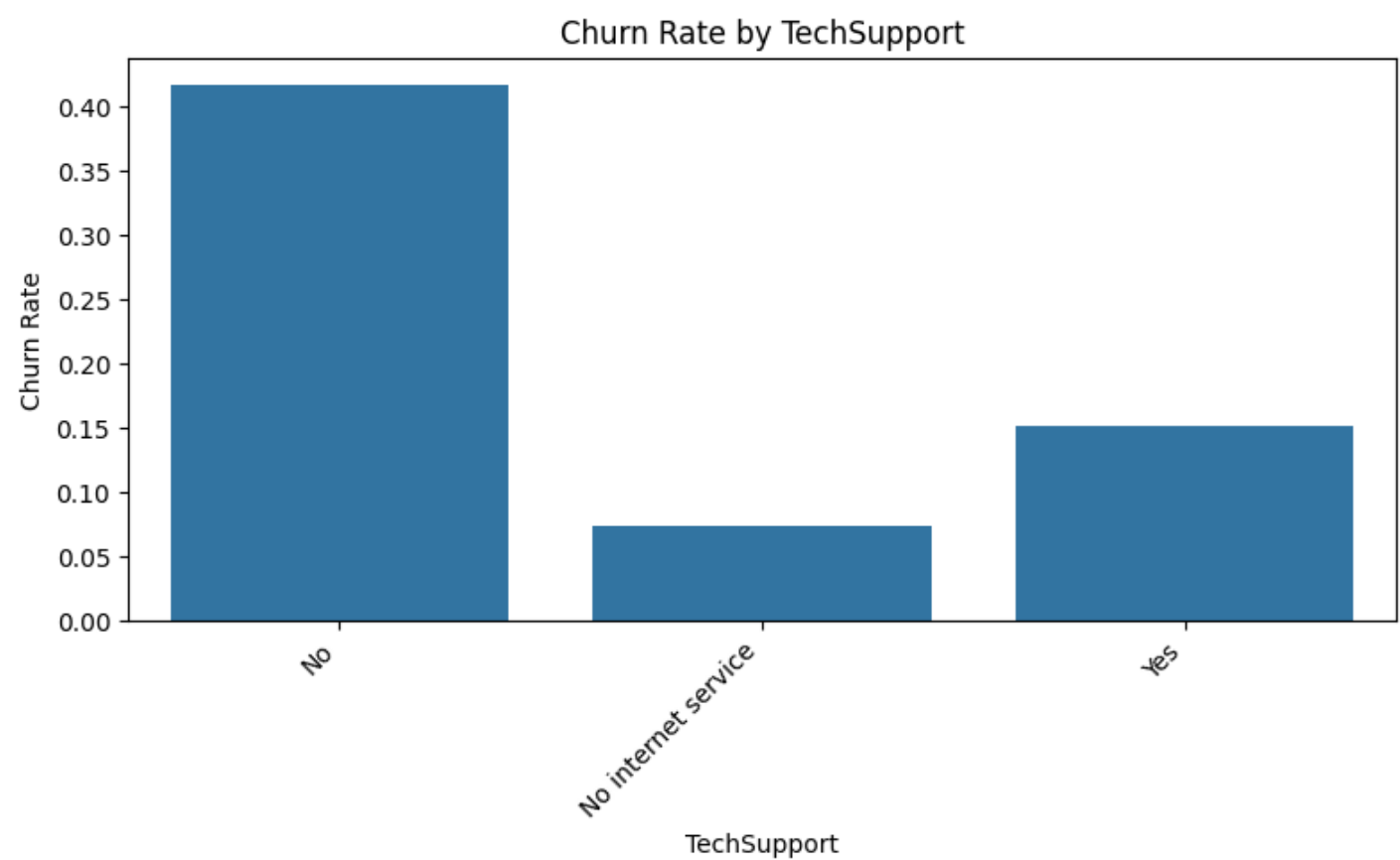
Churn Rate by OnlineBackup



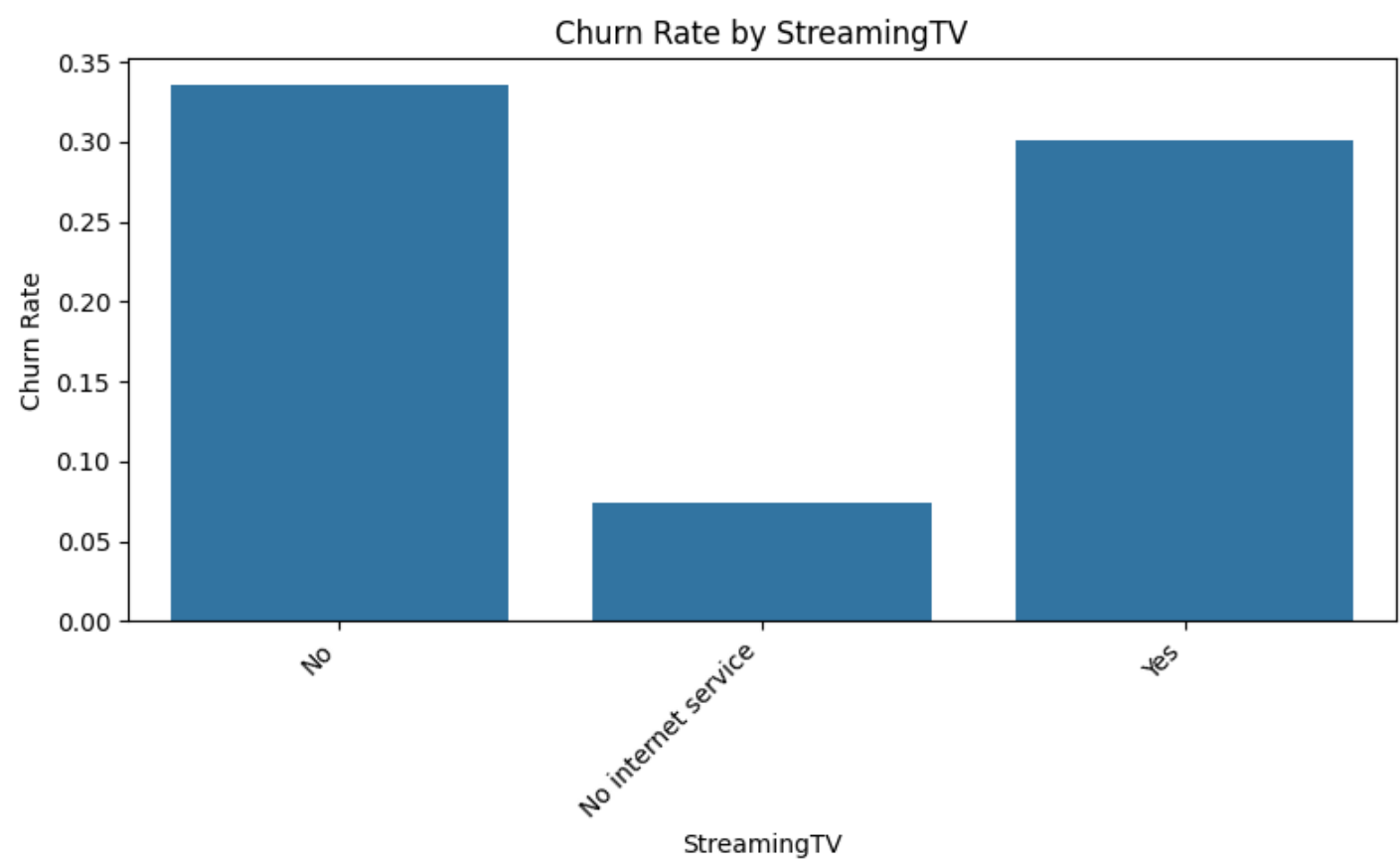
Churn Rate by DeviceProtection



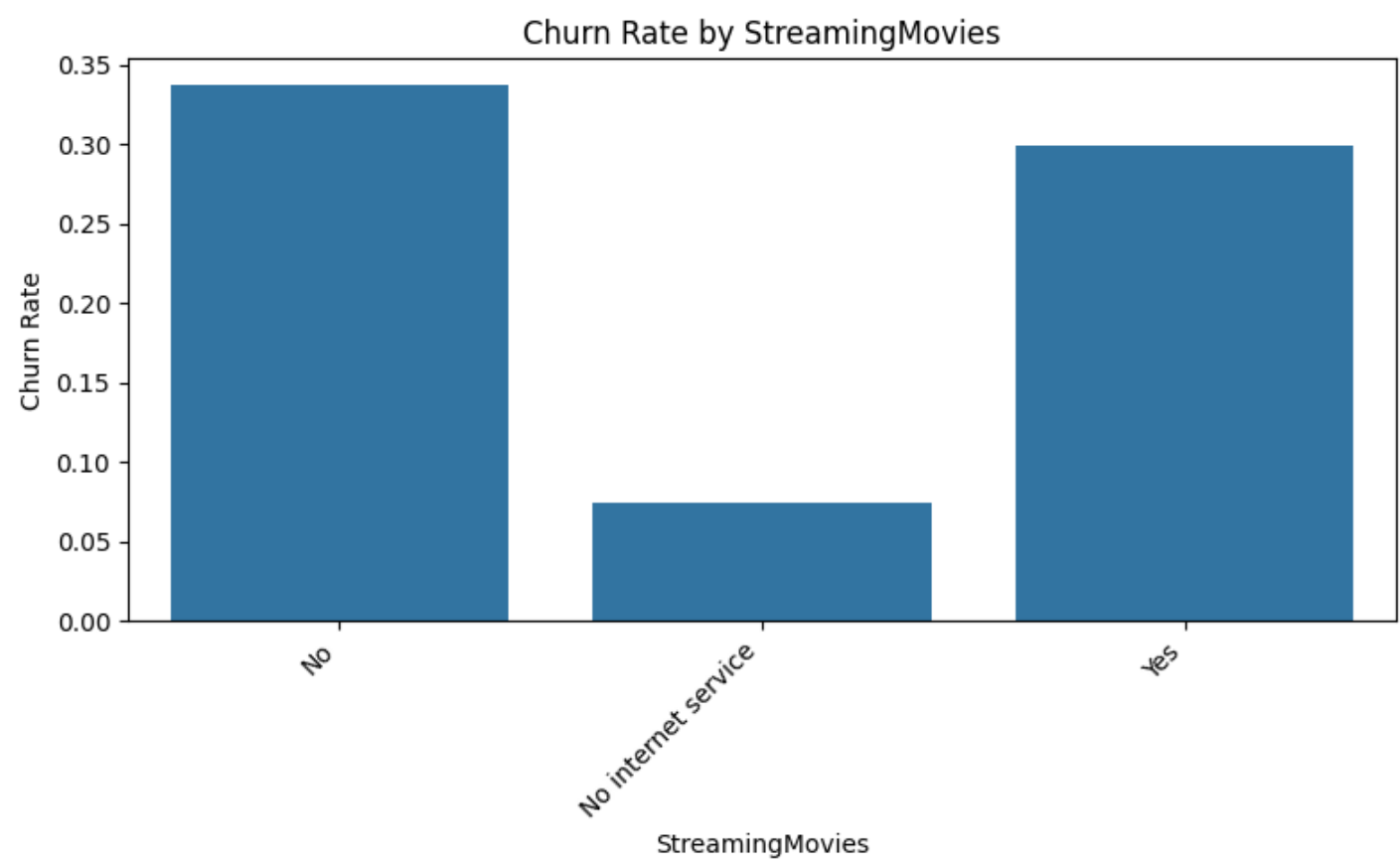
Churn Rate by TechSupport



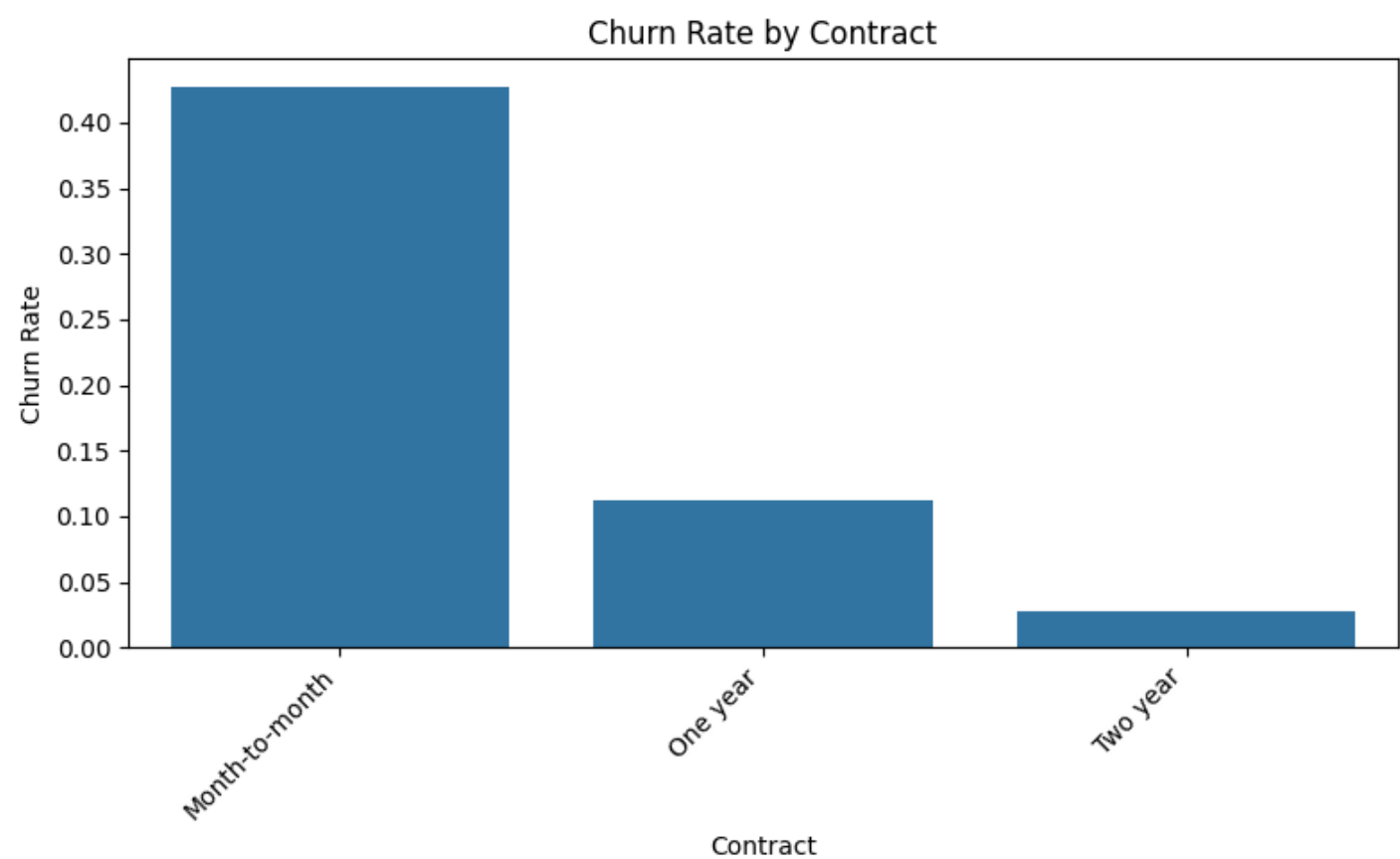
Churn Rate by StreamingTV



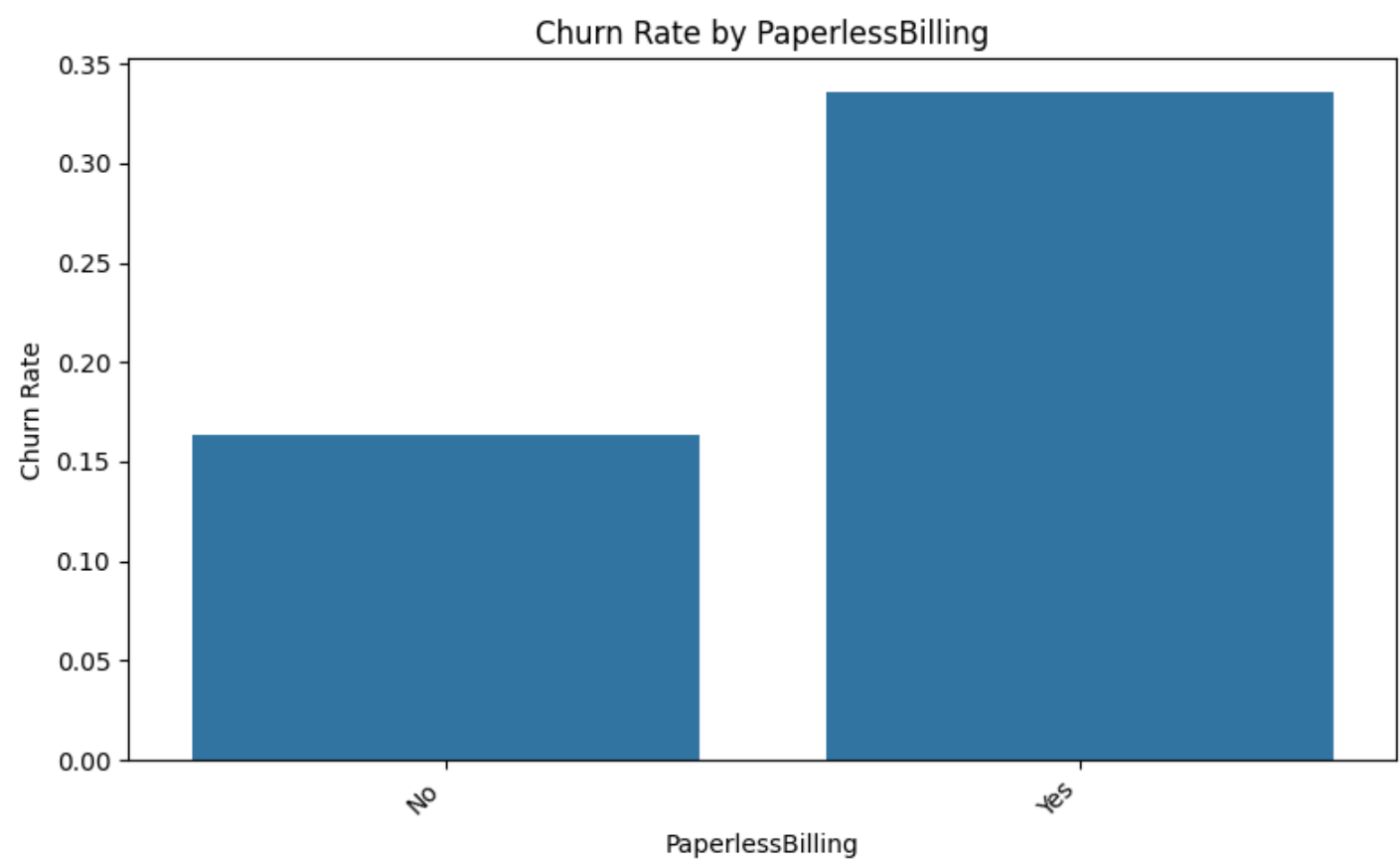
Churn Rate by StreamingMovies



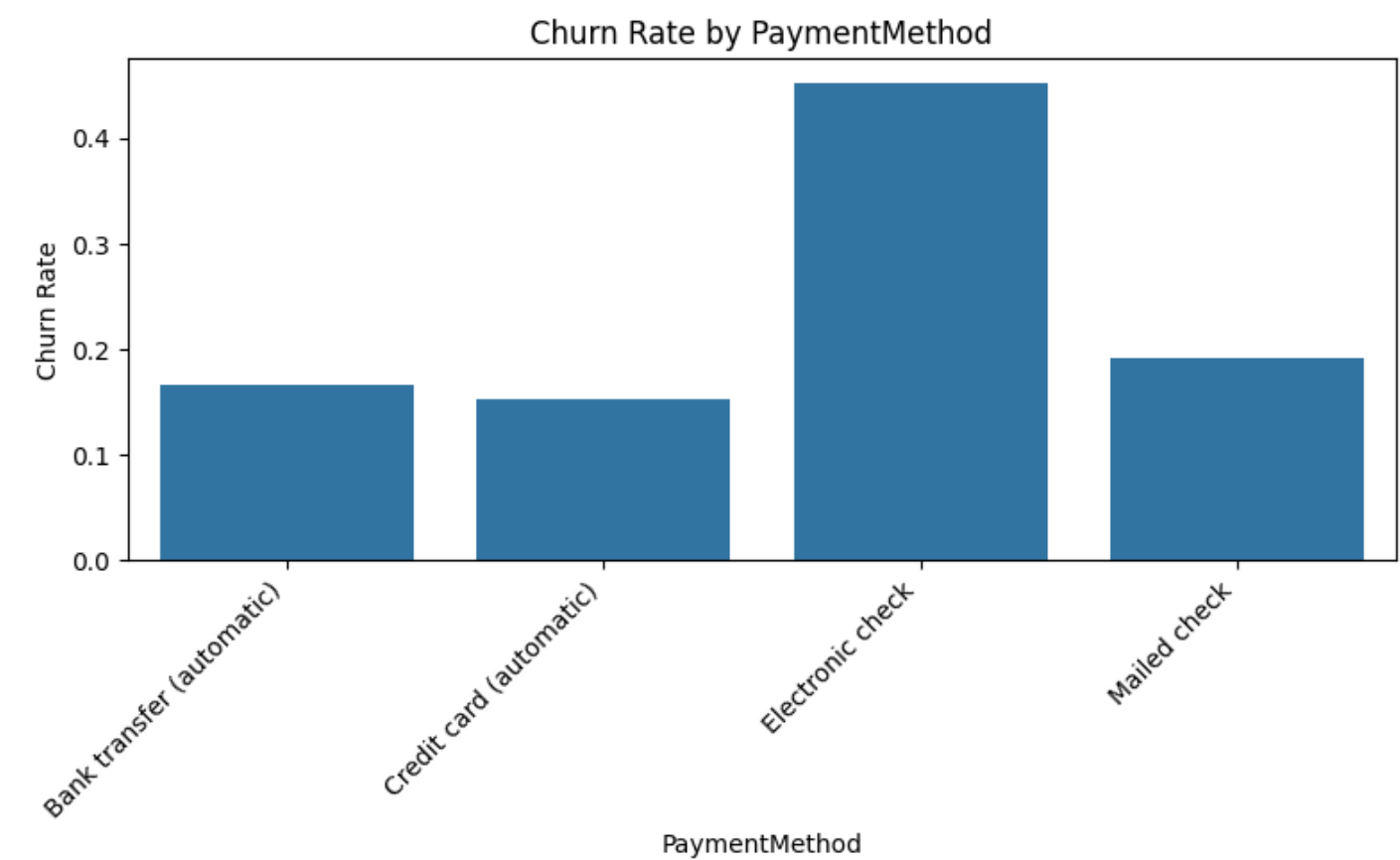
Churn Rate by Contract



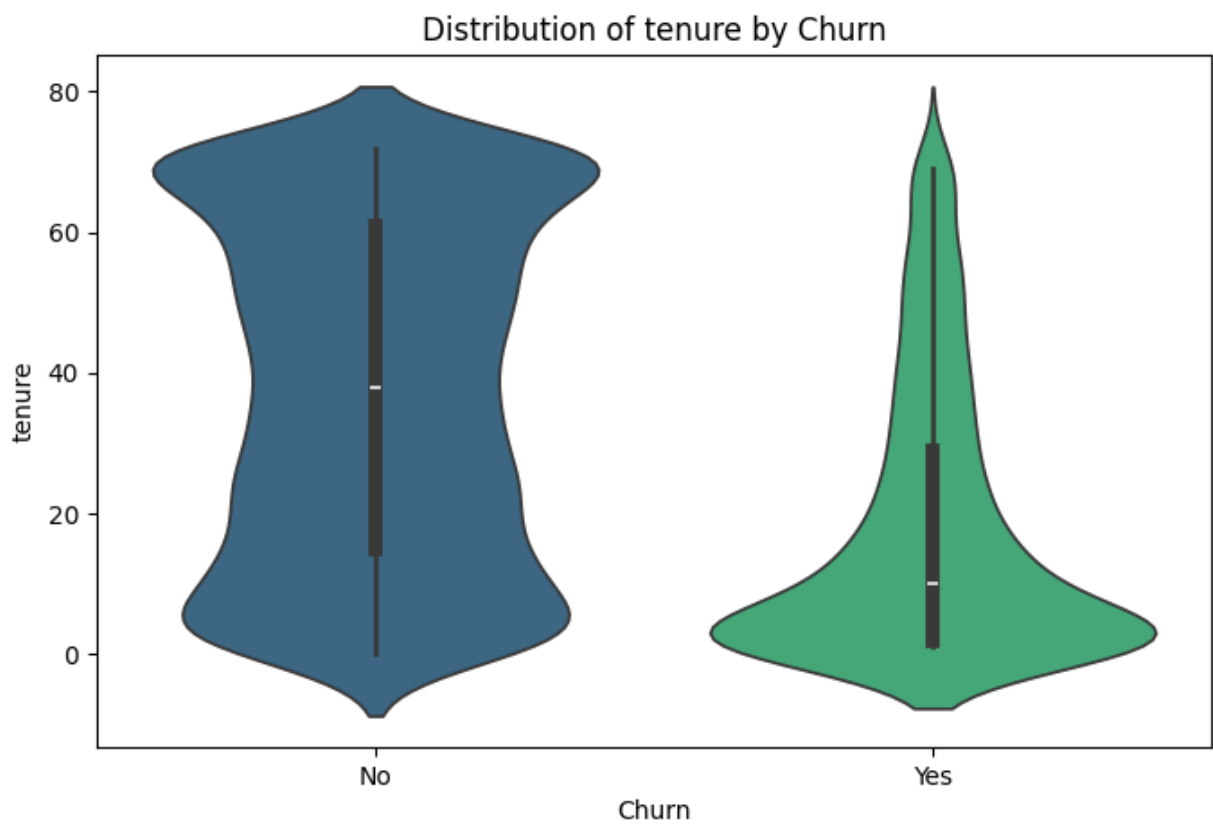
Churn Rate by PaperlessBilling



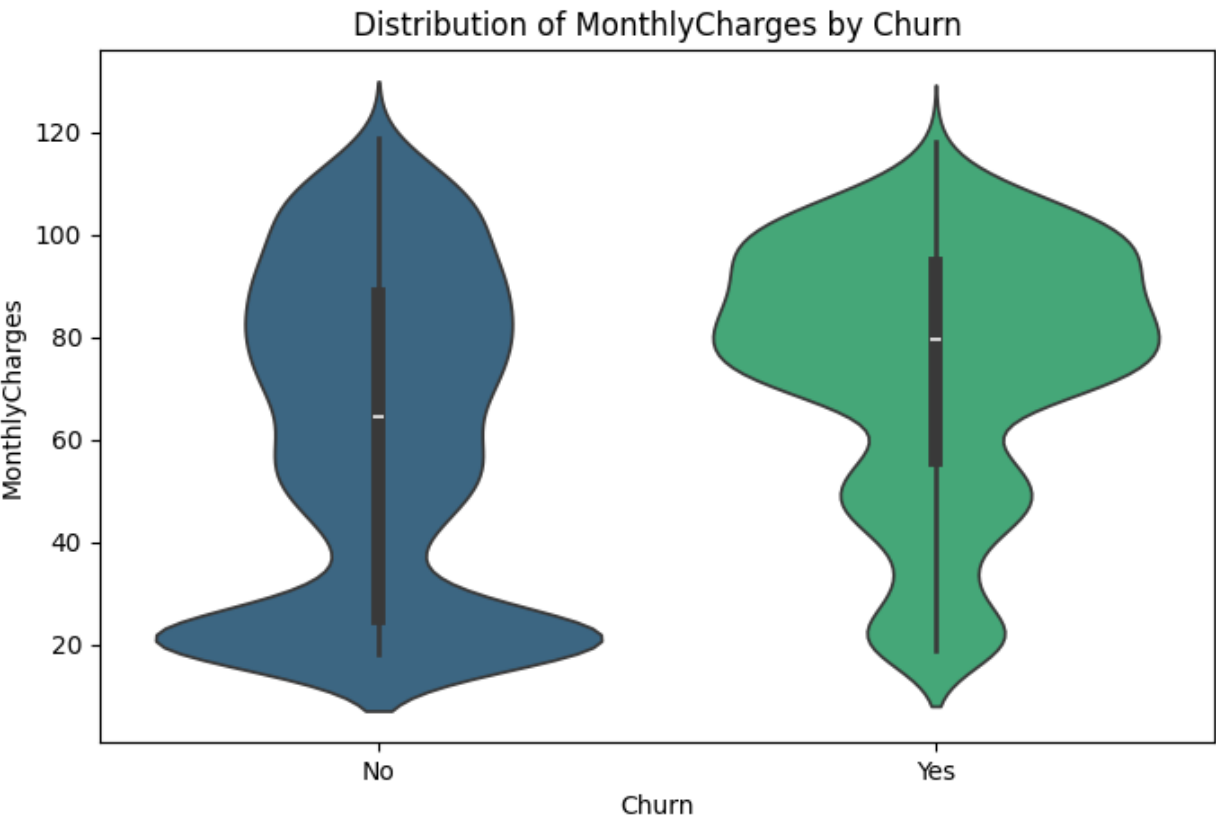
Churn Rate by PaymentMethod



Distribution of tenure by Churn



Distribution of MonthlyCharges by Churn



Distribution of TotalCharges by Churn

